

VIREN VADHVANA

v.vadhvana99@gmail.com | +44 (0)7429 117 628 | LinkedIn | GitHub | Website

PERSONAL STATEMENT

First-Class Computer Science MSci graduate with strong demonstrable technical impact in systems and software at a top hedge fund. Seeking a challenging role to leverage a passion for AI/ML, security, and distributed systems.

EDUCATION

University of Nottingham, UK

September 2021 - July 2026

MSci Computer Science with a Placement Year - First Class (Hons)

- Software Engineering
 - Led an Agile team to deliver a Cyber Threat Intelligence dashboard suggesting ROI-optimised countermeasures
 - Deployed fully tested and documented full-stack solutions using TDD, Git version control, and CI/CD pipelines
 - Authored comprehensive systems specifications and stakeholder requirements, using Figma UI/UX wireframing, ethnographic user research, and heuristic usability analysis to accelerate development and reduce technical debt
- Artificial Intelligence / Machine Learning
 - Trained temporal Graph Neural Networks (tGNNs) to detect illicit Bitcoin transactions on the Elliptic dataset, enriched with external engineered features including market volatility and news sentiment
 - Evaluated ANNs, Kernel SVMs (KSVMs), and XGBoost on complex medical datasets for binary tumor classification, tuning hyperparameters to maximize recall and AUC
 - Engineered an NLP-based flight-booking chatbot, prioritising AI explainability and contextual error handling
- Security
 - Designed and implemented a novel decentralised cryptographic mixing protocol, optimising packet routing and guaranteeing pseudonymity under coordinated adversarial attack models
 - Solved Algorand's non-attribution flaw using Timed VRFs, simulating a reputation-based consensus mechanism, achieving a reduction in malicious behaviour by up to 100%

FULL-TIME EXPERIENCE

Systems Administrator - D. E. Shaw & Co. (London) LLP

June 2024 - June 2025

- Automated global Helpdesk reporting, saving 10+ hours of weekly manual analysis and improving insights and trend visibility for managers, by developing a GenAI (RAG) system with LangChain and LangGraph
- Architected HashiCorp Vault integration with internal tooling, standardising authentication across 10+ databases
- Streamlined L1/L2 troubleshooting and eliminated context switching, consolidating 5+ fragmented REST APIs and CLI tools into a single dashboard, by deploying a unified full-stack platform
- Resolved 500+ production issues, maintaining high SLA adherence and minimal mean time to resolution (MTTR)

PART-TIME EXPERIENCE

Smart Bar IT Assistant - University of Nottingham

September 2022 - July 2026

- Provided L1/L2 hardware, software, and network support and beta-tested patches, achieving 100% user satisfaction

Self-employed Private Tutor - GoStudent

January 2023 - May 2024

- Designed and delivered tailored 1:1 STEM lessons to 10+ college students, helping each to attain top grades

SKILLS

Programming Languages: Python, C, Java, JavaScript, Haskell, SQL/NoSQL, Go, ARM Assembly

Tools and Technologies: Git, Kubernetes, Unix, PowerShell, AWS, Blockchain, Web technologies, LangChain

EXTRACURRICULARS

Tae Kwon Do 2nd Dan Black Belt and TAGB Umpire: coach students and officiate competitions 2011 - present

Musician: grade 8 piano and keyboard player with high distinctions (>90%) in every exam taken 2011 - present

Hackathon Organiser/Volunteer: led events by engaging with sponsors and coordinating logistics 2022 - 2026

Peer Mentor: provided academic and pastoral support to new students at the University of Nottingham 2021 - 2026

Open-Source Assistive Device Designer: led design and manufacture of custom prosthetic leg cover 2023 - 2024